To Create Hive Database

Theory

Apache Hive is a data warehouse software built on top of the Hadoop ecosystem. It is used to store, manage, and analyze large datasets that are stored in the Hadoop Distributed File System (HDFS). Hive provides an SQL-like language called HiveQL (Hive Query Language), which allows users to perform database operations and query large amounts of data without writing complex MapReduce programs.

Hive is mainly used for data warehousing and batch processing. It converts HiveQL queries into execution tasks that can be processed using supported execution engines such as MapReduce, Tez, or Spark. It provides a simple interface for managing structured and semi-structured data.

A Hive Database is a logical collection of tables that helps organize data. It provides a structured way to store and access related datasets. Users can create databases, tables, insert records, retrieve data, and perform various operations using HiveQL commands.

Important Features of Hive

1. SQL-like Query Language: Hive provides HiveQL, which is similar to SQL and makes data querying easier.
2. Data Warehousing: It is designed to store and analyze large volumes of data.
3. Hadoop Integration: Hive works with Hadoop and uses HDFS for distributed data storage.
4. Scalability: It can process large datasets distributed across multiple machines.
5. Schema on Read: The structure of data is applied when the data is read rather than when it is initially stored.
6. Data Management: It supports database and table creation, modification, deletion, and data retrieval.

Steps to Create a Hive Database

Hive provides several commands to create and manage databases.

1. Start Hive:

Open the terminal and start the Hive command-line interface using:

hive

2. Create a Database:

The CREATE DATABASE command is used to create a new database.

```
CREATE DATABASE student_db;
```

This command creates a database named student_db.

3. Display Available Databases:

The SHOW DATABASES command displays all databases available in Hive.

```
SHOW DATABASES;
```

4. Select a Database:

The USE command is used to select the database in which further operations will be performed.

```
USE student_db;
```

5. Create a Table:

The CREATE TABLE command creates a table inside the selected database.

```
CREATE TABLE student (
   id INT,
   name STRING,
   age INT,
   department STRING
);
```

6. Display Tables:

The SHOW TABLES command displays all tables present in the selected database.

```
SHOW TABLES;
```

7. Describe Table Structure:

The DESCRIBE command displays the structure and column details of a table.

DESCRIBE student;

8. Insert and Retrieve Records:

INSERT INTO student

VALUES (1, 'Rahul', 21, 'Computer');

SELECT * FROM student;

The INSERT INTO command adds records, while the SELECT command retrieves the stored records.

Advantages of Hive

1. Simplifies querying of large datasets using HiveQL.
2. Supports distributed storage and processing through Hadoop.
3. Handles structured and semi-structured data.
4. Provides scalability for big data applications.
5. Reduces the need to write complex MapReduce programs.

Conclusion

Apache Hive is an important data warehousing tool in the Hadoop ecosystem. It provides an SQL-like interface to create databases, manage tables, and analyze large datasets. Using HiveQL commands such as CREATE DATABASE, USE, CREATE TABLE, and SELECT, users can efficiently organize and retrieve data for big data analysis.